

Holosynthics: From Code to DNA

Validating the periodic table of digital matter on biological data

Abstract

Holosynthics decomposes any digital artifact into eight irreducible atom types governed by seven universal laws — a 56-element periodic table. This case study extends the proof from seven digital domains to biology. Four adapters decompose DNA sequences, protein 3-D structures, clinical records, and high-throughput sequencing reads — validated on public, peer-cited datasets — into the same table that handles source code and audio. The gene **BRCA1** resolves to the identical element across three biological formats, bringing total cross-domain validation to ten independent formats. Every result is code-verified (166 tests, zero failures) and Ed25519-signed.

Background — seven domains already proven

Prior work established decomposition on text, code, image, audio, PDF, HTTP, and LiDAR, plus a cross-domain proof in which the word "car" resolves to the same Identity element across all seven. The claim under test here: biology — the most complex digital data there is — follows the same table, with no special-casing.

The seven laws — one law, three names

Universality is demonstrated by naming the same law in three registers: the universal science name (the authority), the internal code alias, and the domain-specific biological derivative.

Holosynthics (Science)	Code	Biology (Derivative)
Conservation of Proof	Conservation	Sequence Fidelity
Irreducibility	Distinction	Base Atomicity
Selective Expression	Criticality	Gene Expression
Contextual Resonance	Feedback	Codon Context
Composition	Composition	Molecular Assembly
Inheritance	Network	Genetic Inheritance
Convergence	Complexity	Evolutionary Convergence

Methods — four formats, four adapters

Each adapter implements a deterministic decompose→compose round-trip: raw bytes to typed atoms and back, verified by hash. FASTA reads bases as Symbol atoms positioned in Space and tagged with Identity. PDB reads 3-D coordinates as Number in Space³ — a labeled point cloud. Clinical CSV reads measurements as Number, diagnoses as Boolean, identifiers as Identity. FASTQ reads bases as Symbol and per-base quality as Entropy — confidence built into the table, not bolted on.

Results — real numbers

Adapter	Dataset	Atoms	Key counts	Round-trip
bio.fasta	BRCA1 mRNA (NCBI U14680.1)	7,630	5,711 bp · GC 41.55%	byte-exact hash
bio.pdb	BRCA1/BARD1 (RCSB 1JM7)	575,898	44,268 ATOM records · 14 NMR models	byte-exact

Adapter	Dataset	Atoms	Key counts	Round-trip
bio.fastq	Ebola RNA-seq (SRR1553607)	118 (summary)	203,445 reads	gzip verified
bio.clinical	METABRIC breast cancer	6,612	1,904 patients · 693 features	tabular

Every atom carries a specific element — zero unclassified across all four files. The law axis is exercised narrowly on these four files (a subset of the 56 elements populated); a broader corpus fills more cells. We state this plainly rather than overclaim.

Cross-domain identity — BRCA1

"BRCA1" decomposes to the same element — **Identity · Conservation of Proof** (biology: Sequence Fidelity) — in the FASTA header, the PDB structure, and the clinical spreadsheet. Three formats, three scientific contexts, one element. With the existing seven-domain "car" proof, this is ten independent formats on one table.

Two honest findings

Where the data disagreed with the initial specification, the adapters emit what the data actually is: `brca1` and `brca1_mut` collapse to one Identity atom for the gene, not two; and `overall_survival` is Symbol + Number + Relation (a censored time-to-event with a patient linkage), not a Boolean.

Conclusion

Every digital thing — including the most complex biological data — is made of the same eight atom types under the same seven laws. Biology is not special; it follows the table. Ten formats, one periodic table, cryptographically signed and reproducible.

Verification: `cargo test -p gi-decomposition = 166 passed / 0 failed`; `clippy -D warnings clean`. Ed25519 signature over sha256(canonical proof); pubkey `afb3d04e...74e208`. Data: NCBI, RCSB, EBI/NCBI, METABRIC (public).